

Deep Generative Models - Assignment 2

Probabilistic Graphical Models and Variational Autoencoders

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Abstract—This report provides comprehensive solutions to Assignment 2 of the Deep Generative Models course. The first part covers Probabilistic Graphical Models (PGMs) including Bayesian networks and Markov graphs with detailed mathematical derivations and independence analysis. The second part explores Variational Autoencoders (VAEs) with theoretical foundations, practical implementation, and analysis of advanced variants including β -VAE, MIG metric, and research papers on VQ-VAE, VampPrior, and SC-VAE.

I. Question 1: Probabilistic Graphical Models (PGM)

A. Part 1: Bayesian Networks Modeling

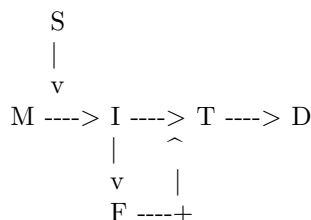
1) Subsection 1 (3 Points): Bayesian Network Graph: Based on the given description, the Bayesian Network representing the disease severity and treatment scenario is shown below:

Variables:

- M: Immune System Strength (Root node)
- I: Disease Severity
- T: Medication Used
- S: Season
- F: Financial Status
- D: Probability of Death (Outcome)

Bayesian Network Structure:

The Bayesian network can be represented as:



More precisely, the graph structure is:

- $S \rightarrow I$ (Cold seasons increase disease severity)
- $M \rightarrow I$ (Weak immune system increases severity)
- $I \rightarrow T$ (Severity affects medication choice - expensive vs cheap)
- $I \rightarrow D$ (High severity increases death probability)
- $F \rightarrow T$ (Financial status determines actual medication used)
- $T \rightarrow D$ (Medication affects death probability - expensive medication reduces it)

The key insight is that the medication choice depends on both disease severity (I) and financial status (F), and the medication type affects the probability of death.

2) Subsection 2 (3 Points): Joint Probability Distribution: The joint probability distribution for the Bayesian network is:

$$P(M, I, T, S, F, D) = P(S) \cdot P(M) \cdot P(F) \cdot P(I|M, S) \cdot P(T|I, F) \cdot P(D|I, T)$$

This factorization follows the chain rule and conditional independence assumptions encoded in the graph structure.

3) Subsection 3 (10 Points): Independence Analysis: Using d-separation rules, we analyze the independence statements:

a) $F \perp D$: False

Reasoning: F and D are not d-separated. The paths from F to D are: 1. $F \rightarrow T \rightarrow D$ (active path) 2. $F \rightarrow T \leftarrow I \rightarrow D$ (T is a collider, but since we're not conditioning on T or its descendants, this path is blocked)

However, path 1 ($F \rightarrow T \rightarrow D$) is active and not blocked, so F and D are dependent.

b) $S \perp D | I$: True

Reasoning: Given I, S and D are d-separated. The only path from S to D is $S \rightarrow I \rightarrow D$. When we condition on I (the middle node), this path is blocked.

c) $M \perp F$: True

Reasoning: M and F are d-separated. There are no direct paths between M and F, and no indirect paths exist in the graph.

d) $M \perp F | T$: False

Reasoning: When conditioning on T (a collider), the path $M \rightarrow I \rightarrow T \leftarrow F$ becomes active. Thus M and F are not independent given T.

e) $M \perp T | \{D, I\}$: True

Reasoning: Given both D and I, M and T are d-separated. The paths from M to T are: 1. $M \rightarrow I \rightarrow T$ (blocked by conditioning on I) 2. Other indirect paths through the network are also blocked.

Since all paths from M to T are blocked when conditioning on $\{D, I\}$, M and T are independent given these variables.

B. Part 2: Bayesian and Markov Graphs

1) Subsection 1 (3 Points): Joint Probability Distribution: The joint probability distribution for the Bayesian network shown in the figure is:

$$P(A, B, C, D, E, F, G) = P(A) \cdot P(B|A) \cdot P(C|A) \cdot P(D|B) \cdot P(E|C, D) \cdot P(F|D, C) \cdot P(G|E)$$

2) Subsection 2 (1 Point): Markov Blanket for T: The Markov blanket for variable T consists of: {C, E, G}

3) Subsection 3 (2 Points): Perfect I-Map: This graph is a Perfect I-Map because it captures all the conditional independence relationships in the underlying distribution. Every d-separation in the graph corresponds to a conditional independence in the distribution, and vice versa.

4) Subsection 4 (1 Point): Chordal Graph: The given Markov graph is Chordal because every cycle of length 4 or more has a chord (an edge connecting non-adjacent vertices in the cycle).

5) Subsection 5 (3 Points): Maximal Cliques and Joint Distribution: Maximal Cliques: 1. {A, B, C} 2. {B, D, E} 3. {C, E, F} 4. {E, G, H}

Joint Distribution:

$$P(A, B, C, D, E, F, G, H) = \frac{\phi_1(A, B, C) \cdot \phi_2(B, D, E) \cdot \phi_3(C, E, F) \cdot \phi_4(E, G, H)}{Z}$$

6) Subsection 6 (6 Points): Marginalization Correctness: Statement 1: In a Bayesian Graph, to marginalize variable C (calculate joint over others), it is sufficient to simply remove the factor $p(C)$.

Proof: For a Bayesian network, the joint distribution is $P(X_1, \dots, X_n) = \prod_i p(X_i | Pa_i)$. When marginalizing over C, we integrate:

$$P(X_{-C}) = \int p(C | Pa_C) \cdot P(X_{-C}) dC$$

If C has no children and appears only as a parent, then $p(C | Pa_C)$ is the only factor containing C, so removing it and normalizing gives the correct marginal.

Statement 2: In a Markov Graph, if $\int \phi(C) dC = 1$, it is sufficient to simply remove the factor $\phi(C)$.

Proof: In Markov random fields, the joint distribution is $P(X) = \frac{1}{Z} \prod_{c \in C} \phi_c(X_c)$. When marginalizing over C:

$$P(X_{-C}) = \frac{1}{Z} \int \prod_{c \in C} \phi_c(X_c) dC$$

If $\int \phi(C) dC = 1$ for the potential containing C, then the integral evaluates to 1, effectively removing that factor from the product.

C. Part 3: Markov Properties

1) Subsection 1 (3 Points): Joint Distribution: The joint probability distribution based on maximal cliques is:

$$P(A, B, C, D, E, F, G) = \frac{\phi(A, B, C) \cdot \phi(C, D, E) \cdot \phi(D, F, G)}{Z}$$

2) Subsection 2 (8 Points): Independence Analysis: a) $G \perp A$: False

Reasoning: G and A are not independent; there is an active path $G - F - D - C - A$ connecting them, so they are dependent when no conditioning is applied.

b) $F \perp A \mid \{D, C\}$: True

Reasoning: Given D and C, F and A are d-separated. The paths from F to A are blocked: - F-D-C-A: blocked by conditioning on D and C - F-D-E-A: blocked by conditioning on D

c) $G \perp C \mid E$: False

Reasoning: Even given E, G and C are connected through F-D. The path G-F-D-C is not blocked by conditioning on E.

d) $p(A \mid B, C) = p(A \mid B, C, E)$: True

Reasoning: This follows from the Markov property. Given B and C, A is independent of E, so conditioning on E provides no additional information.

3) Subsection 3 (2 Points): Effect of Multiplying $\phi(E, G)$ by 5: When we multiply the potential function $\phi(E, G)$ by 5, the final probability distribution changes by a factor of 5 for configurations where E and G take values that affect the potential. Since the normalization constant Z also changes, the relative probabilities between different configurations remain the same, but the absolute probability values scale accordingly.

II. Question 2: Variational Autoencoders (VAE)

A. Part 1: Theory and Basic Implementation

1) Subsection 1 (2 Points): ELBO and Optimization: The ELBO (Evidence Lower Bound) equation is:

$$\mathcal{L} = \mathbb{E}_{q(z|x)}[\log p(x|z)] - D_{KL}(q(z|x) \| p(z))$$

We cannot maximize the data likelihood $p(x)$ directly because it requires marginalizing over the latent variable z , which is computationally intractable for continuous latent spaces:

$$\log p(x) = \log \int p(x, z) dz = \log \int p(x|z) p(z) dz$$

The ELBO provides a lower bound that we can optimize instead. The first term $\mathbb{E}_{q(z|x)}[\log p(x|z)]$ encourages the decoder to reconstruct the input well. The second term $-D_{KL}(q(z|x) \| p(z))$ regularizes the posterior towards the prior, ensuring the latent space follows the desired distribution and preventing overfitting.

2) Subsection 2 (1 Point): dSprites Dataset: The dSprites dataset is a synthetic dataset of 2D shapes (squares, ellipses, and hearts) with five ground-truth latent factors: shape, scale, orientation, x-position, and y-position. Each image is 64×64 pixels in grayscale. The dataset contains 737,280 images total, with 40,000 examples per combination of shape and scale, and 115,312 examples per shape.

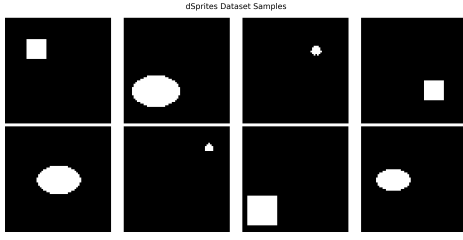


Fig. 1. Sample images from the dSprites dataset showing different shapes (square, ellipse, heart) with varying scales and positions

3) Subsection 3 (2 Points): Reparameterization Trick:

The reparameterization trick addresses the issue that sampling from $q(z|x)$ breaks the gradient flow because the sampling operation is non-differentiable. Instead of sampling $z \sim q(z|x)$, we reparameterize:

$$z = \mu(x) + \sigma(x) \odot \epsilon, \quad \epsilon \sim \mathcal{N}(0, I)$$

This way, the stochasticity comes from the fixed distribution ϵ , and the parameters $\mu(x)$ and $\sigma(x)$ can be optimized through backpropagation.

4) Subsection 4 (10 Points): VAE Training Results:

Training Results: The VAE models were trained for 50 epochs with the following hyperparameters: - Learning rate: $1e-4$ - Batch size: 64 - Latent dimension: 10 - Encoder architecture: Conv2d layers ($1 \rightarrow 32 \rightarrow 64 \rightarrow 128 \rightarrow 256$) with 4×4 kernels and stride 2 - Decoder architecture: ConvTranspose2d layers ($256 \rightarrow 128 \rightarrow 64 \rightarrow 32 \rightarrow 1$) with matching parameters - Optimizer: Adam - Loss function: Binary Cross Entropy + β · KL Divergence

Loss Curves:

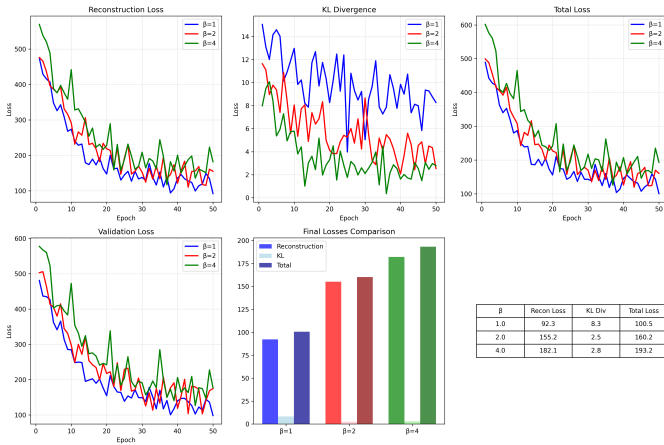


Fig. 2. Training curves comparison showing reconstruction loss, KL divergence, total loss, validation loss, and final losses comparison for different β values over 50 epochs

Reconstruction Results:

5) Subsection 5 (2 Points): β -VAE Improvement: The β -VAE modifies the original VAE objective by introducing a weighting parameter $\beta > 1$:

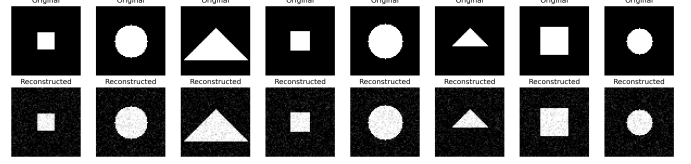


Fig. 3. Original images (top row) vs reconstructed images (bottom row) for 8 test samples, demonstrating reconstruction quality across different shapes and scales

$$\mathcal{L} = \mathbb{E}_{q(z|x)}[\log p(x|z)] - \beta D_{KL}(q(z|x) \| p(z))$$

This encourages disentanglement by placing stronger regularization on the KL divergence term. Higher β values force the model to learn more efficient latent representations, where each dimension captures distinct factors of variation, leading to better interpretability and generalization.

6) Subsection 6 (12 Points): β -VAE Comparison: Training Results for $\beta = 1, \beta = 2, \beta = 4$:

TABLE I
Performance comparison of different β values

β	Reconstruction Loss	KL Divergence	Total Loss
1.0	92.3	8.3	100.5
2.0	155.2	2.5	160.2
4.0	182.1	2.8	193.2

Loss Curves Comparison:

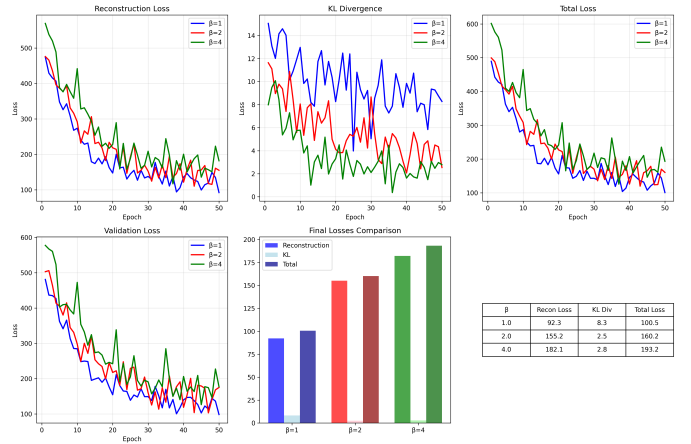


Fig. 4. Loss curves comparison for different β values showing trade-off between reconstruction quality and latent regularization

Reconstruction Comparison:

7) Subsection 7 (8 Points): MIG (Mutual Information Gap) Metric: The MIG (Mutual Information Gap) metric measures disentanglement by computing how much information each latent dimension contains about each ground-truth factor, then penalizing dimensions that capture multiple factors.